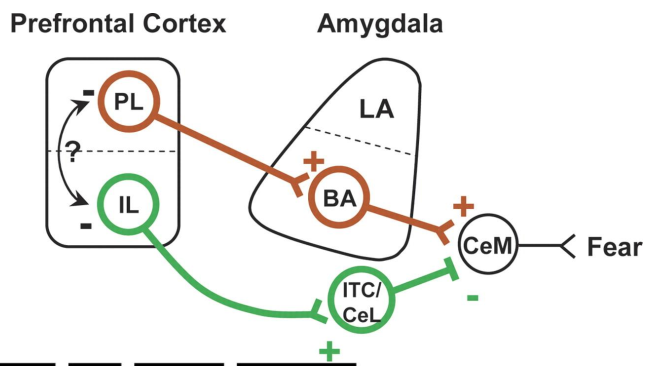
Build-A-Zombie

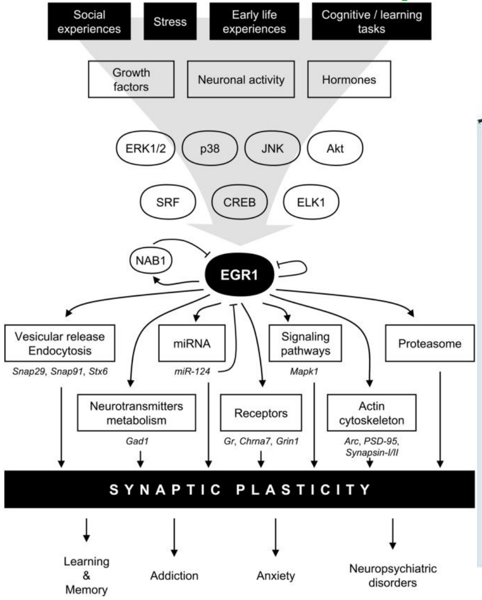
Feeding, Cognition, & Aggression: Group #1
The BioWeapon: SARS-COV-Z
"Ceaseless Fury"

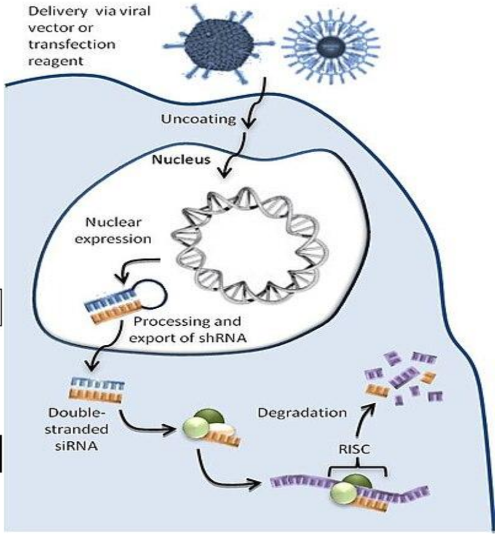
Fearless - AMYGDALA

1. Mutant **Ca2+** channel encoded by SARS-COV-Z stimulates neurons, including the GABAergic ITC (Intercalated) cells in the infralimbic Cortex which **inhibits** the amygdala

2. Short-hairpin RNA in SARS-COV-Z has tight RNA hairpins that forms double-stranded siRNA in the cell which targets the Arc gene, EGR1 gene, and GluN2A genes (subunit of NMDA receptor) which are critical in the amygdala for fear conditioning. The siRNA binds to these DNA sequences and SILENCES them = no fear learning







Genes silenced by shRNA

The **Arc protein** is in excitatory synapses: regulates AMPA receptor endocytosis in the amygdala - prevents the glutamate receptor endocytosis - increased glutamate transmission which is excitatory and thus long-term potentiation for long-term fear learning.

GluN2A codes for a subunit of the NMDA receptor for glutamate - increased glutamate transmission in the amygdala, which is **excitatory**.

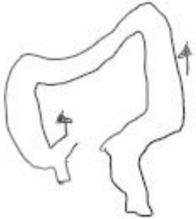
ERG1 (Early Growth Response 1) - Synaptic formation and plasticity so it will be involved in fear learning. Increase of ERG1 in the lateral nucleus of the amygdala right after fear conditioning.Decreasing ERG1 will decrease the ability to form new connections

Memory - THIAMINE DEFICIENCY

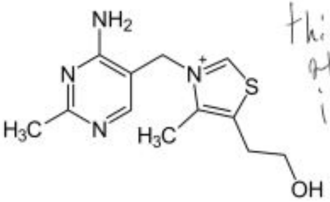
SARS-CoV-Z -intestinal cell damage: The ileum and colon showed comparatively higher expression of ACE2, indicating that intestinal epithelial cells are susceptible to infection. By damaging them, SARS-CoV-Z can infect absorptive enterocytes and cause diarrhea and decreasing their absorption.

By decreasing the intestinal absorption, there is **less absorption of thiamine (Vitamin B1)**.

- In Wernicke-Korsakoff syndrome (caused by alcoholism), severe retrograde amnesia is linked to thiamine deficiency, which is why these patients are usually given thiamine as treatment. Thiamine is an important molecule in processing glucose in the brain, including in the hippocampus for long-term memory consolidation.
- Thiamine can be reabsorbed by the kidneys
    - However, high epinephrine released by the adrenal gland by our virus **increases** blood pressure which signals to the posterior pituitary gland to **inhibit** antidiuretic hormone, which will **increase** urination and so thiamine will NOT be reabsorbed



↑ ACE2 expression damages

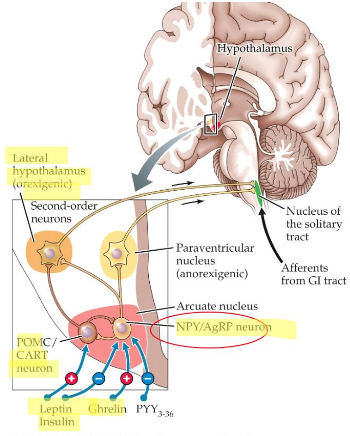


thiamine absorption ↓ in ileum + colon

Kidney absorption damaged by
↑ epinephrine
↳ ↑ urination

Feeding - MORE FEEDING

In previous lab, we increased T3 and T4 production → increased fat breakdown. There is LESS leptin being released by fat tissue → less activation of POMC/CART → less inhibition of lateral hypothalamus, which is OREXIGENIC = MORE FEEDING. Also intestines are damaged = less absorption of food = need to eat more.



BIOLOGICAL PSYCHOLOGY 7e, Figure 13.23 (Part 2)